

Quantitative Mean-Reversion Strategy Research Project

Executive Summary

This project investigates a systematic mean-reversion trading strategy using historical financial market data and Python-based quantitative analysis. The primary objective was to explore whether deviations from rolling statistical baselines could be used to generate trading signals and whether those signals exhibited predictive characteristics under varying market conditions.

The strategy employs rolling z-score analysis to identify statistically significant price deviations from a moving average. Trading signals are generated when prices move beyond predefined z-score thresholds, under the assumption that extreme deviations may subsequently revert toward their historical mean. The research framework includes exploratory backtesting, parameter sensitivity analysis, volatility-regime analysis, Monte Carlo simulation, and robustness testing across multiple assets and market environments.

Although the project was developed as a research-oriented simulation rather than a production trading system, it provides a practical framework for evaluating systematic trading concepts and demonstrates core quantitative finance methodologies.

Project Objectives

The project was designed to address the following research questions:

1. Can rolling statistical deviations be used to generate meaningful trading signals?
 2. How sensitive is strategy performance to parameter selection?
 3. Does strategy performance remain stable under different volatility regimes?
 4. How robust are the results under randomized return sequences?
 5. Can the strategy generalize across multiple asset classes?
 6. What risks emerge from drawdowns, tail events, and model uncertainty?
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Methodology

Data Collection

Historical daily market data were downloaded using Yahoo Finance through the Python `yfinance` package.

Typical assets analyzed included:

- SPY (S&P 500 ETF)
- QQQ (NASDAQ-100 ETF)
- GLD (Gold ETF)
- BTC-USD (Bitcoin)
- ETH-USD (Ethereum)

The analysis period spanned multiple market cycles, including periods of expansion, crisis, recovery, and elevated volatility.

Return Calculation

Daily logarithmic returns r_t were computed as:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

where:

- P_t is the current closing price
- P_{t-1} is the previous closing price

Logarithmic returns were chosen because they are additive through time and commonly used in quantitative finance research.

Mean-Reversion Signal Generation

The strategy computes rolling statistics using a fixed lookback window.

For each trading day:

Rolling Mean: μ_t

Rolling Standard Deviation: σ_t

The rolling z-score z_t is then calculated as:

$$z_t = \frac{P_t - \mu_t}{\sigma_t}$$

Trading rules:

Long Signal: $z_t < -\text{Threshold}$

Short Signal: $z_t > \text{Threshold}$

Neutral: otherwise

The underlying hypothesis is that large deviations from historical norms may subsequently revert toward equilibrium levels.

Backtesting Framework

Signals are converted into positions using a one-period lag to avoid look-ahead bias.

Strategy returns are computed as:

Strategy Return = Position $\times$ Asset Return

An equity curve is generated using cumulative compounding of strategy returns.

Benchmark performance is evaluated using a buy-and-hold portfolio over the same period.

Exploratory Data Analysis

Several statistical characteristics of asset returns were examined:

- Mean return
- Volatility
- Skewness
- Kurtosis

Distributional analysis revealed the presence of:

- Non-normal return distributions
- Fat tails
- Volatility clustering
- Asymmetric downside risk

These observations highlight the limitations of simple Gaussian assumptions commonly used in theoretical finance models.

Volatility Regime Analysis

Market conditions were segmented into:

- Low-volatility regimes
- Medium-volatility regimes
- High-volatility regimes

Performance was analyzed separately within each regime.

This allowed evaluation of whether the mean-reversion effect strengthened or weakened during periods of market stress and uncertainty.

Parameter Sensitivity Analysis

A parameter grid search was performed across combinations of:

- Rolling window lengths
- Z-score thresholds

For each parameter combination:

- Signals were generated
- Backtests were executed
- Risk-adjusted performance metrics were calculated

Results were visualized using heatmaps.

The objective was to determine whether profitability depended on a narrow parameter set or remained relatively stable across a broad range of configurations.

A strategy exhibiting stable performance across multiple parameter choices is generally considered more robust and less susceptible to overfitting.

Monte Carlo Simulation

Monte Carlo simulations were performed using bootstrap resampling of historical strategy returns.

The process involved:

1. Randomly sampling historical returns with replacement.
2. Constructing simulated return paths.
3. Generating simulated equity curves.
4. Repeating the process thousands of times.

Outputs included:

- Simulated portfolio trajectories
- Distribution of terminal portfolio values
- Confidence intervals
- Tail-risk estimates

This analysis provided insight into uncertainty surrounding future strategy outcomes.

Risk Analysis

Several risk metrics were evaluated:

Sharpe Ratio

Measures risk-adjusted return:

Sharpe Ratio = Mean Excess Return / Return Volatility

Higher values indicate more favorable risk-adjusted performance.

Maximum Drawdown

Measures the largest peak-to-trough decline in portfolio value.

This metric quantifies downside exposure experienced by an investor during adverse periods.

Value-at-Risk (VaR)

Estimated the portfolio loss threshold not expected to be exceeded at a specified confidence level.

Conditional Value-at-Risk (CVaR)

Measures the expected loss conditional on losses exceeding the VaR threshold.

CVaR provides a more informative estimate of extreme tail risk than VaR alone.

Robustness Testing

Several robustness tests were conducted to assess model reliability.

Walk-Forward Validation

The dataset was divided into:

- Training period
- Validation period
- Testing period

Parameters were optimized only on training data and subsequently evaluated on unseen data.

This procedure helps reduce overfitting and better approximates real-world deployment conditions.

Cross-Asset Validation

The strategy was tested across multiple asset classes.

Consistent performance across assets suggests the possibility of a more general market phenomenon rather than a dataset-specific anomaly.

Stress Testing

Performance was evaluated during periods of elevated market stress, including crisis environments characterized by rapid price movements and increased volatility.

Special attention was given to drawdown behavior and recovery dynamics.

Results Summary

The research produced several key findings:

1. Mean-reversion signals occasionally identified statistically extreme market conditions.
 2. Performance varied significantly across parameter selections.
 3. Risk-adjusted returns were generally more informative than raw returns.
 4. Monte Carlo simulations revealed substantial dispersion in possible future outcomes.
 5. Drawdown analysis highlighted the importance of risk controls.
 6. Robustness testing demonstrated that some parameter combinations generalized better than others.
 7. Cross-asset testing suggested that strategy effectiveness was dependent on market structure and asset characteristics.
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Limitations

Several limitations should be considered:

- No transaction costs were included.
- No slippage assumptions were modeled.
- Market impact was ignored.
- Execution latency was not considered.
- Position sizing remained simplistic.
- The strategy relied on historical price information only.
- Future market conditions may differ materially from historical observations.

Consequently, results should be interpreted as research findings rather than evidence of deployable profitability.

Conclusions

This project demonstrates a complete quantitative research workflow for evaluating systematic trading strategies. Beginning with data acquisition and exploratory analysis, the framework progresses through signal generation, backtesting, parameter sensitivity analysis, Monte Carlo simulation, risk assessment, and robustness testing.

Beyond evaluating a specific mean-reversion concept, the project emphasizes broader quantitative finance principles including statistical validation, uncertainty quantification, risk management, and model robustness. The resulting framework serves as a foundation for future research into systematic trading strategies and provides practical experience with the tools and methodologies commonly used in quantitative finance and algorithmic trading research.

Technologies Used

- Python
 - NumPy
 - Pandas
 - Matplotlib
 - SciPy
 - Seaborn
 - yFinance
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Skills Demonstrated

- Quantitative Research
- Financial Time Series Analysis
- Statistical Modeling
- Backtesting Methodology
- Monte Carlo Simulation
- Risk Analysis
- Data Visualization
- Python Development
- Systematic Trading Research
- Performance Evaluation
- Model Robustness Testing